

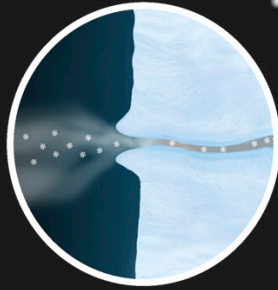
Mars

Explorations of Mars found water ice at its poles and there may be liquid water under the surface. However, Mars has very little atmosphere and is blasted by radiation. If there is life, it probably lies beneath the surface.



Enceladus and Europa

These two moons are thought to have liquid water under their icy surfaces. Both have hot interiors, so organisms could be living around underwater hydrothermal vents similar to those on Earth.



Io

This is one of the few moons with an atmosphere. It is volcanically active and has a hot inner core. There is evidence of complex chemicals, but any life would have to cope with lethal radiation from Jupiter.



Titan

Titan has the most potential for supporting life. Its atmosphere contains amino acids that are the basis for life on Earth. The conditions are similar to those of a young Earth, although there is no liquid water.



LITTLE GREEN MEN

In movies, aliens are usually made to look like humans but with big almond-shaped eyes and no hair, or as strange creatures with three heads and five arms. But there is no guarantee that they will look like that. They may resemble some of the weird and wonderful things that live on Earth because certain body shapes and structures are ideal for certain environments: torpedo shapes are best for swimming, paired legs are good for walking, wings are necessary for flying, and eyes are useful for detecting light. Evolution on other planets is as likely to have produced as many strange organisms as it has on Earth.



This is no alien—it is a creature called *Opabinia* that lived on Earth 500 million years ago!

LOW
GRAVITY



IN DIFFICULT CONDITIONS

If there is life on another planet it probably has to contend with different physical conditions than Earth. It may have a different gravity, day length, temperature, or atmosphere, which would have an effect on the organism's body shape, way of moving, energy needs, and life cycle. For example, life on planets with high gravity would be very low to the ground to survive the crushing pressure of the atmosphere. Imagine a world of short, squat people and plants. The opposite would happen with low gravity—plants would grow very tall and animals would have to be tall to reach them.

HIGH
GRAVITY



Out of this world

